1. What are Logical Functions?

Logical functions in Excel help us make decisions based on certain conditions. They check whether a condition is TRUE or FALSE and return the appropriate result.

Example: If a student's marks are 40 or above, display “Pass”; otherwise, display “Fail”.

2. IF Function

The IF function checks a condition and returns one result if the condition is TRUE and another result if it is FALSE.

Syntax

=IF(logical_test, value_if_true, value_if_false)

Argument	Meaning
logical_test	The condition to check
value_if_true	Result if the condition is TRUE
value_if_false	Result if the condition is FALSE

Example: A student passes if their marks are at least 40.

=IF(A2>=40,"Pass","Fail")

Marks	Result
75	Pass
40	Pass
25	Fail

Here, A2>=40 checks whether marks are greater than or equal to 40.

3. Comparison Operators

Comparison operators help us create conditions in Excel.

Operator	Meaning	Example
>	Greater than	A2>50

<	Less than	A2<50
>=	Greater than or equal to	A2>=40
<=	Less than or equal to	A2<=40
=	Equal to	A2=100
<>	Not equal to	A2<>0

4. AND Function

The AND function returns TRUE only when all conditions are TRUE.

Syntax

=AND(condition1, condition2)

Example: An employee qualifies for a bonus if they have at least 3 years of experience AND earn at least $85,000.

=AND(A2>=3,B2>=85000)

Experience	**Salary**	**Result**
5 years	120000	TRUE
2 years	135000	FALSE
4 years	75000	FALSE

Both conditions must be satisfied for the result to be TRUE.

5. OR Function

The OR function returns TRUE when at least one condition is TRUE.

Syntax

=OR(condition1, condition2)

Example: An employee qualifies if they have at least 5 years of experience OR earn at least $100,000.

=OR(A2>=5,B2>=100000)

If either condition is satisfied, the result is TRUE. Both conditions can also be TRUE.

6. NOT Function

The NOT function reverses a logical result.

- TRUE becomes FALSE.
- FALSE becomes TRUE.

Example:

=NOT(A2="Data Analyst")

If A2 contains Data Analyst, the result is FALSE. If A2 contains a different job title, the result is TRUE.

7. Combining IF with AND and OR

We can combine logical functions with IF to display meaningful messages instead of TRUE or FALSE.

IF + AND

Requirement: An employee receives a bonus only if experience is at least 3 years AND salary is at least $85,000.

=IF(AND(A2>=3,B2>=85000),"Bonus","No Bonus")

The formula returns Bonus only when both conditions are TRUE.

IF + OR

Requirement: An employee is eligible if experience is at least 5 years OR salary is at least $100,000.

=IF(OR(A2>=5,B2>=100000),"Eligible","Not Eligible")

The formula returns Eligible if either condition is TRUE.

Difference between AND and OR

AND	**OR**
All conditions must be TRUE.	At least one condition must be TRUE.
=AND(TRUE,TRUE) → TRUE	=OR(TRUE,FALSE) → TRUE
=AND(TRUE,FALSE) → FALSE	=OR(FALSE,FALSE) → FALSE

8. Nested IF Function

A Nested IF means using one IF function inside another IF function.

It is useful when there are more than two possible outcomes.

Syntax

=IF(condition1,result1,IF(condition2,result2,result3))

Example 1: Salary classification

Suppose the annual salary is in cell B2.

- Salary of $100,000 or above → High Salary
- Salary of $70,000 or above but below $100,000 → Medium Salary
- Salary below $70,000 → Low Salary

Formula:

=IF(B2>=100000,"High Salary",IF(B2>=70000,"Medium Salary","Low Salary"))

Annual Salary	**Result**
120000	High Salary
85000	Medium Salary
50000	Low Salary

How it works:

1. Excel first checks whether salary is at least $100,000.
2. If TRUE, it displays High Salary.
3. If FALSE, Excel checks whether salary is at least $70,000.
4. If TRUE, it displays Medium Salary; otherwise, it displays Low Salary.

Example 2: Student grading

Suppose marks are in cell A2.

Marks	**Grade**
90 or above	A
75–89	B
50–74	C
Below 50	Fail

Formula:

=IF(A2>=90,"A",IF(A2>=75,"B",IF(A2>=50,"C","Fail")))

Excel checks each condition in order and returns the result corresponding to the first TRUE condition.

9. Nested IF with AND and OR

Logical functions can also be combined to solve more complex problems.

Example: A job is marked as Desired if its title is Data Analyst OR Data Scientist, AND its annual salary is at least $85,000.

Suppose:

- A2 contains the job title.
- B2 contains the annual salary.

=IF(AND(OR(A2="Data Analyst",A2="Data Scientist"),B2>=85000),"Desired","Not Desired")

How it works:

1. OR checks whether the job title matches either of the two roles.
2. AND checks whether the job title condition and salary condition are both satisfied.
3. IF returns Desired if both requirements are met; otherwise, it returns Not Desired.